

QUANTUM OPTIMIZATION FOR DISTRIBUTED ORDER MANAGEMENT

Technical Project Report

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Quantum Optimization for Distributed Order Management

EXECUTIVE SUMMARY

Distributed Order Management (DOM) is the process of deciding which distribution center should fulfill each customer order while considering inventory availability, transportation cost, warehouse capacity, and business value. As the number of orders and fulfillment locations increases, the number of possible assignments grows exponentially, making the problem suitable for combinatorial optimization.

This project investigates a hybrid approach combining classical optimization, quantum-inspired optimization, and quantum optimization techniques for distributed order management. The study uses a real-world-inspired dataset containing 614 orders and 1,110 SKUs. To develop a tractable optimization formulation, 359 focus orders were identified, with 93 single-SKU focus orders selected for the main optimization experiments across 8 distribution centers. This resulted in a reduced formulation containing 744 binary order-to-distribution-center decision variables.

The optimization problem was formulated using binary decision variables, an objective function incorporating order value, shipping cost, and penalties for unfulfilled orders, together with order-assignment and capacity constraints. An Integer Linear Programming (ILP) model was used as the exact reference solution. Several alternative approaches were evaluated, including Greedy Assignment, Random Sampling, Simulated Annealing, Tabu Search, Steepest Descent, and Path-Integral Quantum Monte Carlo (PIQMC).

The ILP approach achieved an objective value of approximately 447,822 and served as the reference for comparison. Steepest Descent achieved approximately 444,071, corresponding to 99.2% of the ILP objective, with a runtime of about 0.08 seconds. A separate QAOA experiment was performed on a small 3-order, 2-distribution-center instance containing 6 binary variables. QAOA achieved an objective value of approximately 191,974 compared with a brute-force optimum of approximately 256,412, resulting in an optimality gap of 25.1%.

The results demonstrate that different optimization approaches exhibit different trade-offs between solution quality, runtime, and scalability. The QAOA experiment also provides a practical demonstration of mapping a business optimization problem into a quantum optimization framework, while the results do not establish a quantum advantage for the problem size tested. The project therefore provides a foundation for studying hybrid classical, quantum-inspired, and quantum approaches to large-scale distributed order management.

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1. INTRODUCTION AND BUSINESS UNDERSTANDING

1.1 Distributed Order Management

Distributed Order Management (DOM) is the process of deciding which distribution center should fulfill each customer order. In a distributed supply network, the same product may be available at multiple distribution centers, while inventory levels, warehouse capacities, transportation costs, and order priorities may differ. DOM therefore helps determine an appropriate fulfillment location for each order while considering these operational factors.

In this project, the focus is on assigning customer orders to available distribution centers in a way that improves the overall value of fulfillment. The system considers the value obtained from fulfilling an order, the shipping cost associated with the selected distribution center, and a penalty when an order remains unfulfilled.

1.2 Problem Statement

In a distributed order management environment, customer orders may be fulfilled by multiple distribution centers with different inventory levels, capacities, and transportation costs. The challenge is to determine which distribution center should fulfill each order while satisfying operational constraints and achieving the best overall business outcome.

A simple assignment strategy may select the nearest or default distribution center without considering the complete set of available alternatives. However, when several distribution centers can fulfill the same order, the choice of fulfillment location can affect shipping cost, order value, warehouse utilization, and the number of orders that remain unfulfilled.

Therefore, this project formulates distributed order management as a combinatorial optimization problem. The objective is to select feasible order-to-distribution-center assignments that maximize the overall fulfillment value while accounting for shipping costs, unfulfilled-order penalties, inventory availability, and distribution-center capacity.

1.3 Why Optimization is required

The distributed order management problem becomes challenging when multiple orders can be fulfilled by multiple distribution centers. For each order, there may be several possible fulfillment choices, and every choice can affect inventory usage, transportation cost, warehouse capacity, and the overall value of the solution.

As the number of orders and distribution centers increases, the number of possible assignment combinations grows exponentially. Evaluating every possible combination becomes impractical for larger instances. This makes the problem a suitable application of combinatorial optimization techniques.

Optimization methods can systematically search for feasible and high-value assignments while respecting operational constraints. In this project, different classical, quantum-inspired, and quantum optimization approaches are investigated to understand their solution quality, computational cost, and scalability for the distributed order management problem.

1.4 Project Objectives

The main objective of this project is to formulate and solve the distributed order management problem as an optimization problem and to investigate the applicability of classical, quantum-inspired, and quantum optimization techniques.

The specific objectives are:

- Formulate the order-to-distribution-center assignment problem using binary decision variables.
- Maximize the overall fulfillment value while accounting for shipping costs and penalties for unfulfilled orders.
- Incorporate operational constraints such as one-distribution-center-per-order, inventory availability, and distribution-center capacity.
- Develop a QUBO formulation suitable for quantum optimization.
- Compare classical optimization methods with quantum-inspired approaches.
- Implement and evaluate QAOA on a small problem instance.
- Analyze solution quality, runtime, and scalability across the different approaches.
- Identify the practical limitations and future opportunities for applying quantum optimization to larger distributed order management problems.

2. DATA UNDERSTANDING

2.1 Dataset Overview

The project uses an order-management dataset containing information about customer orders, products, distribution centers, shipping costs, inventory availability, and capacity-related information. The dataset was examined to identify orders that could benefit from reassignment to alternative distribution centers.

The complete dataset contains 614 orders and 1,110 SKUs. From these orders, 359 were identified as focus orders based on inventory availability conditions. To construct a tractable optimization model, 93 single-SKU focus orders were selected for the main experiments. The model considers 8 distribution centers, resulting in 744 binary order-to-distribution-center assignment variables.

2.1.1 Source Data and Key Fields

The project used order-level and SKU-level data associated with the Distributed Order Management problem. The order-level data contains information related to order identification, default and recommended distribution centers, requested and expected delivery dates, ordered quantities, revenue, shipping cost, penalties, distance, lead time, and fulfillment-related measures.

The provided DOM formulation identifies important input information such as SKU, default distribution center, and PGI date. These inputs are used to evaluate whether an order can be fulfilled from the default or an alternative distribution center.

For the optimization experiment, the data was transformed into a reduced order-to-DC decision model. This allowed the project to focus on the assignment decision while keeping the optimization problem computationally tractable.

Table 1. Dataset and optimization scope

Data Item	
Total Orders	614
Total SKUs	1,110
Distribution Centers	8
Focus Orders	359
Single-SKU Focus Orders	93
Binary Decision Variables	744

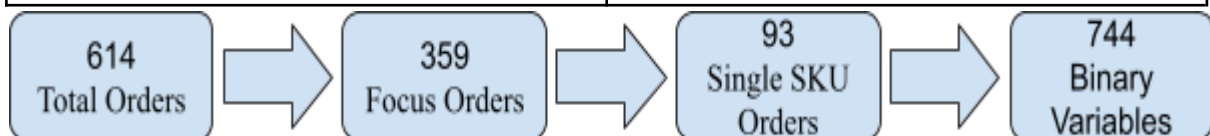


Figure 1. Dataset reduction and optimization-model scope

2.2 Focus-Order Selection

The complete dataset contains 614 orders. Not all orders require alternative fulfillment decisions, so the data was filtered to identify focus orders that required further consideration based on inventory availability. This filtering resulted in 359 focus orders.

For the main optimization experiments, a further selection was made to use single-SKU focus orders. This reduced the optimization instance to 93 orders. The selection was made to keep the mathematical formulation tractable while retaining the essential order-to-distribution-center assignment problem.

Each of the 93 selected orders was evaluated against the 8 distribution centers considered in the model. Therefore, the resulting assignment formulation contains $93 \times 8 = 744$ binary decision variables, where each variable represents whether a particular order is assigned to a particular distribution center.

The reduction from the complete order-level dataset to the optimization subset was performed for computational tractability. The provided business formulation contains more detailed operational rules, while this project focuses on the core order-to-distribution-center assignment decision. Therefore, the optimization dataset should be interpreted as a reduced experimental representation rather than a complete reproduction of the production DOM system.

Table 2. Order-selection stages used in the optimization study

Stage	Orders	Purpose
Complete dataset	614	All available orders
Focus Orders	359	Orders requiring further consideration
Single SKU focus orders	93	Orders used in the main optimization

$$\text{Binary Variables} = 93 \text{ Orders} \times 8 \text{ DCs} = 744$$

2.3 Data Preparation

The data preparation process converts the available order and operational data into a structured optimization input. First, orders are examined for inventory availability to identify focus orders. The focus orders are then filtered to obtain the single-SKU subset used in the main optimization experiment. For each selected order, feasible distribution centers are identified and the corresponding shipping costs and capacity information are prepared. These inputs are then used to construct the binary order-to-distribution-center assignment model containing 744 decision variables.

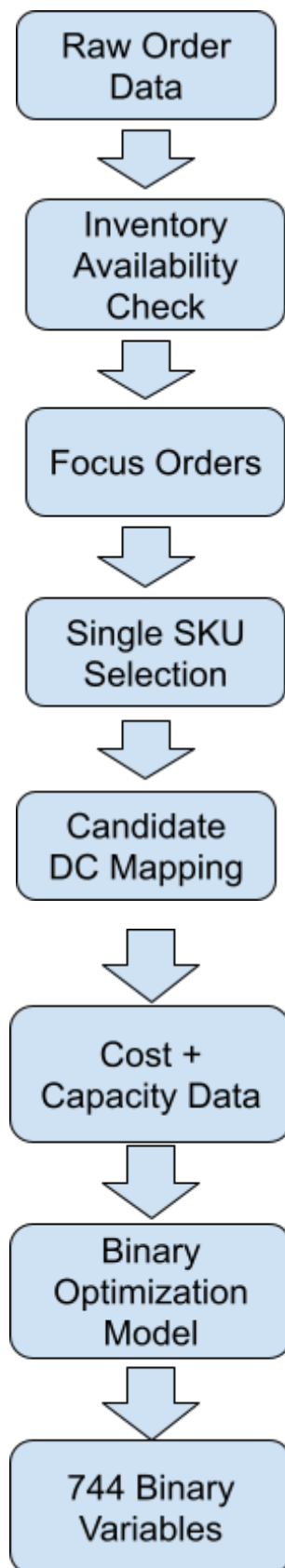


Figure 2. Data preparation and optimization-model construction workflow

3. MATHEMATICAL FORMULATION

3.1 Decision Variables

The optimization model represents each possible order-to-distribution-center assignment using a binary decision variable. For every selected order and distribution center, the variable indicates whether that distribution center is chosen to fulfill the order.

$$x_{\{o, d\}} \in \{0, 1\}$$

where $x_{\{o, d\}}=1$ if order o is assigned to distribution center d , and $x_{\{o, d\}}=0$ otherwise.

3.2 Objective Function

The objective function represents the overall business value of the fulfillment decisions. It combines the value of fulfilling an order with its associated shipping cost. A penalty is also included for orders that remain unfulfilled. Therefore, the model seeks to maximize the net fulfillment value while discouraging unnecessary unfulfilled orders.

$$\left[\max \sum_{o,d} (value_o - shipping\ cost_{o,d}) x_{o,d} - \sum_o penalty_o (1 - \sum_d x_{o,d}) \right]$$

Here, $value_o$ represents the value of order o , $shippingCost_{\{o,d\}}$ represents the transportation cost when order o is fulfilled from distribution center d , and $penalty_o$ represents the business impact of leaving an order unfulfilled. The binary variable $x_{\{o,d\}}$ determines whether the order is assigned to the corresponding distribution center.

A higher objective value indicates a more favorable fulfillment plan under the defined business assumptions.

3.3 Constraints

The optimization model is subject to constraints that ensure the resulting fulfillment plan is operationally feasible. The main constraints considered are the assignment constraint, inventory-related capacity constraints, and distribution-center throughput constraints.

One Distribution Center per Order

Each order can be assigned to at most one distribution center. This prevents the same order from being simultaneously assigned to multiple locations.

$$\sum_d x_{o,d} \leq 1$$

Inventory Capacity

Each distribution center can fulfill an order only when sufficient inventory is available. The model therefore limits assignments according to the available inventory at each distribution center.

$$\sum_o x_{o,d} \leq Capacity_d$$

Distribution Center Throughput

Each distribution center has a limited operational capacity for processing orders. The model ensures that the number of assigned orders does not exceed the modeled throughput capacity of the distribution center.

$$\sum_o x_{o,d} \leq \text{Throughput}_d$$

Binary Decision Constraint

Each assignment variable can take only two values: 1 if the order is assigned to the distribution center and 0 otherwise.

$$x_{o,d} \in \{0, 1\}$$

These constraints ensure that the generated fulfillment plan remains within the defined assignment and operational limits. The resulting constrained optimization problem is then transformed into a QUBO formulation for the quantum optimization approach.

3.3.1 Relation to the Provided DOM Business Formulation

The optimization formulation used in this project was derived from the provided Distributed Order Management (DOM) business formulation. The reference formulation considers order fulfillment across distribution centers using factors such as inventory availability, fulfillment rate, transportation and shipping cost, case-pick and pallet-pick requirements, dock availability, and delivery lead time.

The reference formulation also specifies that an order should be diverted only when the alternative distribution center provides at least a 5% improvement in fill rate compared with the default distribution center. Penalty logic is used to represent the business impact of fulfillment decisions.

For this project, the detailed operational formulation was simplified into a binary order-to-DC assignment model so that it could be studied using classical optimization, quantum-inspired methods, and QAOA. The simplification was necessary to create a tractable mathematical model while retaining the main decision of selecting a suitable distribution center for each order.

3.4 QUBO Formulation

The constrained optimization problem is converted into a Quadratic Unconstrained Binary Optimization (QUBO) formulation to make it suitable for quantum optimization algorithms. QUBO uses binary variables and incorporates constraints into the objective function through penalty terms.

$$\text{QUBO} = \text{Objective} + \text{Penalty Terms}$$

For this project, the binary order-to-distribution-center variables remain the same as in the original formulation. The assignment and capacity requirements are incorporated into the QUBO using penalty terms. The resulting QUBO can then be transformed into an Ising Hamiltonian, which provides the input representation for QAOA.

4. OPTIMIZATION APPROACHES

4.1 Integer Linear Programming (ILP) Baseline

Integer Linear Programming (ILP) was used as the primary classical optimization baseline. The binary order-to-distribution-center variables and the defined objective function and constraints were provided to an ILP solver. The resulting solution serves as a reference for evaluating the performance of the other optimization approaches.

In this project, the ILP model was solved using **PuLP with the CBC solver**. Since the objective value obtained from ILP provides the reference solution for the reduced 93-order problem, the results of heuristic and quantum-inspired methods can be compared against it.

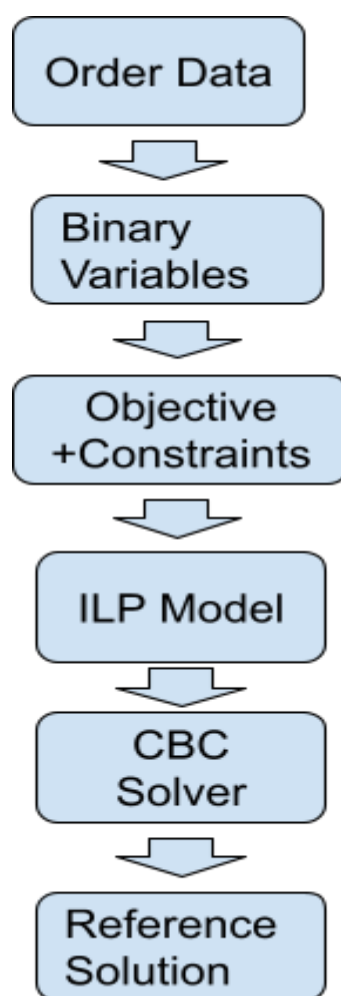


Figure 4. ILP-based classical optimization workflow

4.2 Greedy Assignment

The greedy approach assigns orders sequentially by selecting a locally favorable distribution center based on the available fulfillment options and objective contribution. It provides a simple heuristic solution and serves as a fast baseline for comparison with more advanced optimization methods.

4.3 Random Sampling

Random Sampling is a simple baseline optimization approach in which different feasible order-to-DC assignments are generated randomly. Each assignment is evaluated using the project objective function, and the best solution found among the sampled assignments is selected.

In this project, Random Sampling was used as a baseline to understand how randomly generated solutions compare with more structured optimization methods such as ILP, Steepest Descent, and Tabu Search.

Because the method explores solutions randomly, its performance can vary depending on the number of samples and the random seed. It is computationally simple and fast, but it does not systematically search for the optimal solution.

4.4 Simulated Annealing

Simulated Annealing is a stochastic local-search optimization method inspired by the cooling process of a physical system. Starting from an initial solution, the algorithm explores neighboring solutions and can temporarily accept solutions with lower objective values to escape local optima. As the search progresses, the probability of accepting such solutions decreases.

In this project, Simulated Annealing was applied to the binary order-to-distribution-center assignment problem and evaluated using the same objective function as the other approaches.

4.5 Tabu Search

Tabu Search is a local-search optimization method that uses a memory mechanism to avoid repeatedly exploring recently visited solutions. Starting from an initial feasible solution, the algorithm evaluates neighboring solutions and maintains a tabu list of recent moves. This helps the search explore different regions of the solution space and reduces the possibility of becoming trapped in a local optimum.

In this project, Tabu Search was applied to the order-to-distribution-center assignment problem and its objective value was compared with the ILP baseline and other optimization approaches.

4.6 Steepest Descent

Steepest Descent is a local-search method that repeatedly selects the best available neighboring solution. Starting from an initial solution, the algorithm evaluates possible changes to the current assignment and moves to the neighbor that provides the greatest improvement in the objective value. The process continues until no improving neighboring solution is found.

In this project, Steepest Descent was used to obtain a fast locally optimized solution and compare its performance with the exact ILP baseline and other heuristic methods.

4.7 Path-Integral Quantum Monte Carlo (PIQMC)

Path-Integral Quantum Monte Carlo (PIQMC) is a quantum-inspired classical optimization method based on ideas from quantum statistical mechanics. It represents the optimization problem using a classical probabilistic simulation that incorporates quantum-inspired transitions between candidate solutions.

In this project, PIQMC was applied to the QUBO-based formulation to explore promising binary assignment configurations. The method was evaluated using the same objective function and constraints used for the other optimization approaches.

Although PIQMC is inspired by quantum-mechanical principles, the implementation used in this project is a classical computational method and does not execute on quantum hardware.

Table 3. Optimization approaches evaluated in the project

Approach	Category	Purpose
ILP	Classical exact	Reference Solution
Greedy	Classical heuristic	Fast baseline
Random Sampling	Stochastic	Simple baseline
Simulated Annealing	Metaheuristic	Global search
Tabu Search	Metaheuristic	Guided local search
Steepest Descent	Local Search	Fast improvement
PIQMC	Quantum-inspired	QUBO optimization

5. QUANTUM OPTIMIZATION APPROACH

The quantum optimization stage converts the binary optimization problem into a form suitable for quantum algorithms. The QUBO formulation is first mapped to an Ising Hamiltonian and then used to construct a Quantum Approximate Optimization Algorithm (QAOA) circuit. Because current quantum hardware and simulation resources limit the size of practical experiments, QAOA was evaluated on a small six-variable instance derived from the project problem.

5.1 QUBO to Ising Mapping

QUBO represents the optimization problem using binary variables with values 0 or 1. Quantum algorithms such as QAOA commonly use the Ising representation, where variables are represented using spin values of -1 and $+1$. Therefore, the QUBO variables are transformed into Ising variables before constructing the quantum cost Hamiltonian.

$$x_i = \frac{1+s_i}{2}$$

Here, x_i is the original binary variable and s_i is the corresponding Ising spin variable. Substituting this relationship into the QUBO objective produces an Ising Hamiltonian containing linear and quadratic terms.

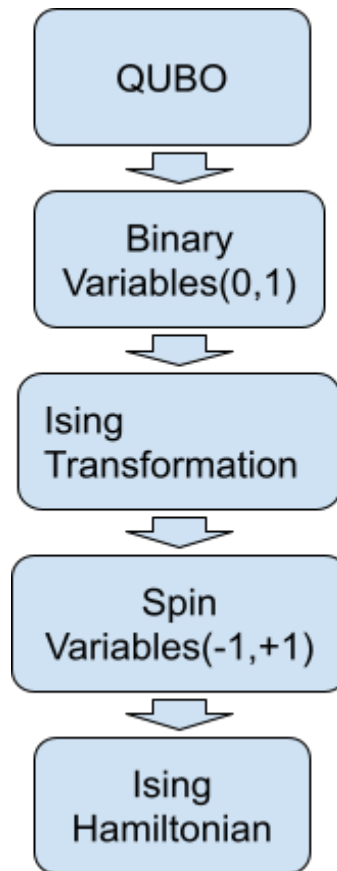


Figure 5. Conversion of the QUBO formulation into an Ising representation

5.2 Quantum Approximate Optimization Algorithm (QAOA)

The Quantum Approximate Optimization Algorithm (QAOA) is a hybrid quantum-classical algorithm designed for combinatorial optimization problems. QAOA uses a parameterized quantum circuit to explore candidate solutions, while a classical optimizer adjusts the circuit parameters to improve the objective value.

QAOA uses two main components: a **cost Hamiltonian**, which represents the optimization objective, and a **mixer Hamiltonian**, which enables exploration of different candidate states. These components are applied repeatedly in layers, controlled by parameters that are optimized classically.

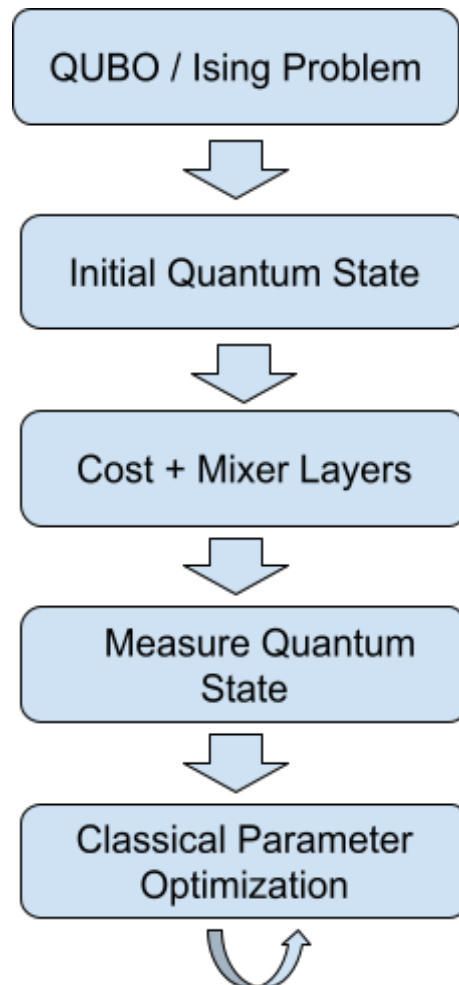


Figure 6. Hybrid quantum-classical QAOA workflow

In this project, QAOA was implemented using Qiskit Aer simulation. Due to the computational cost of simulating larger circuits, the quantum experiment was performed on a small instance containing **3 orders and 2 distribution centers**, resulting in **6 binary variables and 6 qubits**. A QAOA depth of **$p = 2$** was used for the experiment.

The six-qubit experiment was used to investigate the behavior of QAOA on a representative small instance. It was not intended to demonstrate quantum advantage over the full 744-variable optimization problem.

5.3 QAOA Implementation

The QAOA implementation follows a hybrid quantum-classical workflow. First, the selected optimization problem is represented as a QUBO and converted into an Ising-based cost Hamiltonian. The corresponding quantum circuit is then constructed with an initial superposition state followed by alternating cost and mixer operations.

The circuit parameters are optimized using a classical optimization procedure. After parameter optimization, the quantum circuit is measured to obtain candidate binary solutions. These solutions are decoded back into order-to-distribution-center assignments and evaluated using the original project objective function.

For the experimental evaluation, a small instance containing **3 orders and 2 distribution centers** was selected. This produced **6 binary variables**, which were mapped to **6 qubits**. QAOA was executed with $p = 2$, meaning that two alternating cost-and-mixer layers were used. The experiment was performed using the **Qiskit Aer simulator** rather than physical quantum hardware.

744 variables → full reduced optimization model

6 qubits → QAOA experimental subset

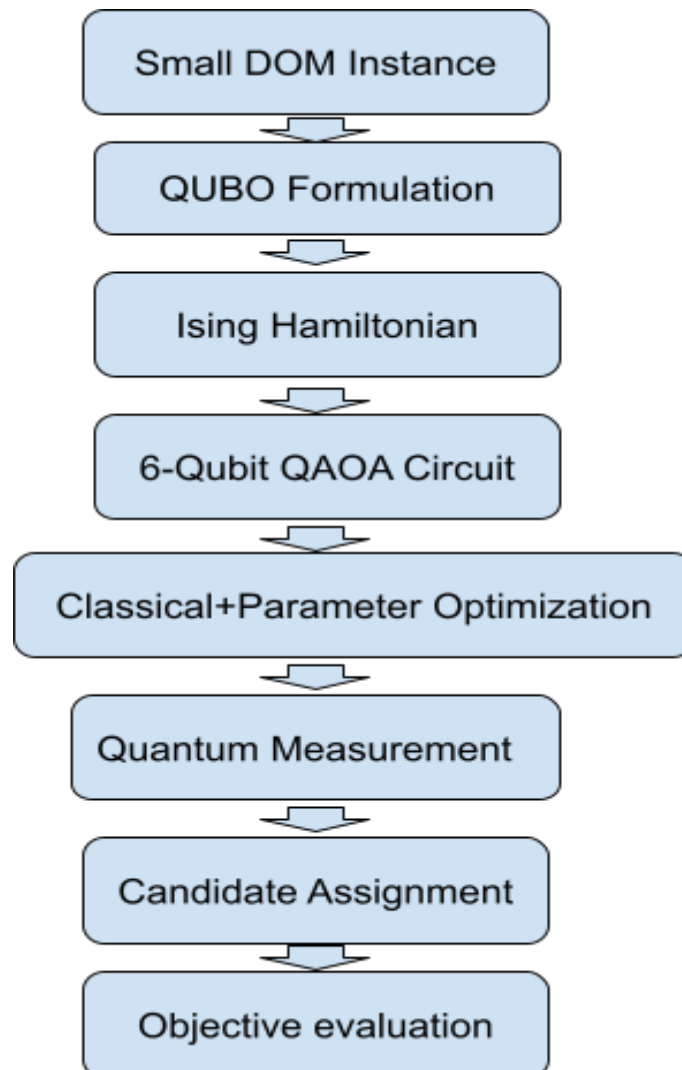


Figure 7. QAOA implementation workflow for the small DOM instance

The complete reduced optimization model contains 744 binary decision variables for 93 orders across 8 distribution centers. Due to the computational requirements of quantum simulation, the QAOA experiment was performed on a smaller instance containing 6 binary variables, corresponding to 6 qubits. Thus, the 6-qubit experiment represents a small experimental subset rather than the full 744-variable problem.

5.4 Small-Instance Experiment

To evaluate the behavior of QAOA, a small instance of the distributed order management problem was selected. The instance contains **3 orders and 2 distribution centers**, resulting in **6 binary decision variables and 6 qubits**. The experiment was executed with **QAOA depth $p=2$** using the Qiskit Aer simulator.

Since the instance contains only six binary variables, all possible binary configurations can be evaluated through brute-force enumeration. This provides an exact reference against which the QAOA result can be compared.

Table 4. Results of the six-qubit QAOA experiment

Metric	Result
Number of variables/qubits	6
QAOA depth	$p=2$
Method	QAOA
Simulator	Qiskit Aer

The brute-force search achieved an objective value of approximately **256,412**, while the QAOA experiment obtained approximately **191,974**. This corresponds to an optimality gap of approximately **25.1%**. The result shows that QAOA was able to produce a feasible candidate solution for the small instance, but it did not reach the exact optimum in this experiment.

QAOA vs Brute Force Objective Value

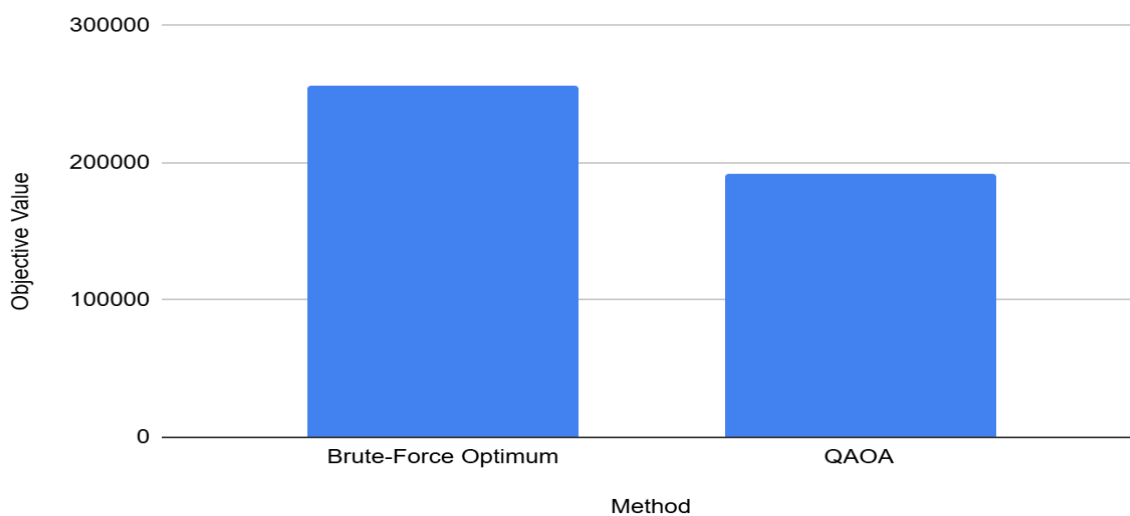


Figure 8. Comparison of brute-force optimum and QAOA objective value

6. EXPERIMENTAL RESULTS

6.1 Classical and Quantum-Inspired Comparison

The optimization approaches were evaluated using the reduced problem containing 93 single-SKU focus orders and 8 distribution centers. The methods were compared using objective value, fulfillment rate, and computational runtime. The ILP solution was used as the reference for evaluating the other approaches.

Table 5. Comparison of optimization approaches

Method	Objective Value	Fill Rate	Run time(s)
ILP	447,822	53%	<2
Steepest Descent	444,071	53%	0.08
Tabu Search	439,188	53%	100.2
Simulated Annealing	369,752	71%	3.2
PIQMC	358,764	69%	2.2
Random Sampling	-8,616	2%	0.003
Greedy	412,060	89%	0.01
Default Assignment	229,995	14%	0.01

Objective Value Comparison

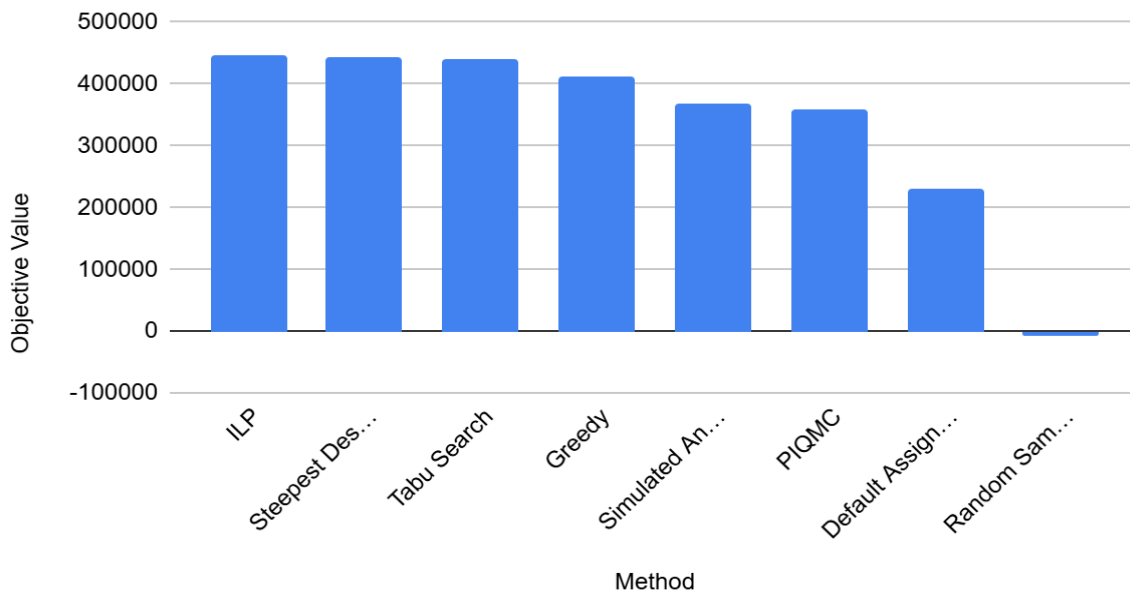


Figure 9. Objective value comparison across optimization approaches

The ILP approach produced the highest objective value among the evaluated methods and was used as the reference solution. Steepest Descent and Tabu Search produced objective values close to the ILP result, while the other heuristic and quantum-inspired approaches produced lower objective values. The differences reflect the search strategies and trade-offs used by the individual methods.

Fill Rate Comparison

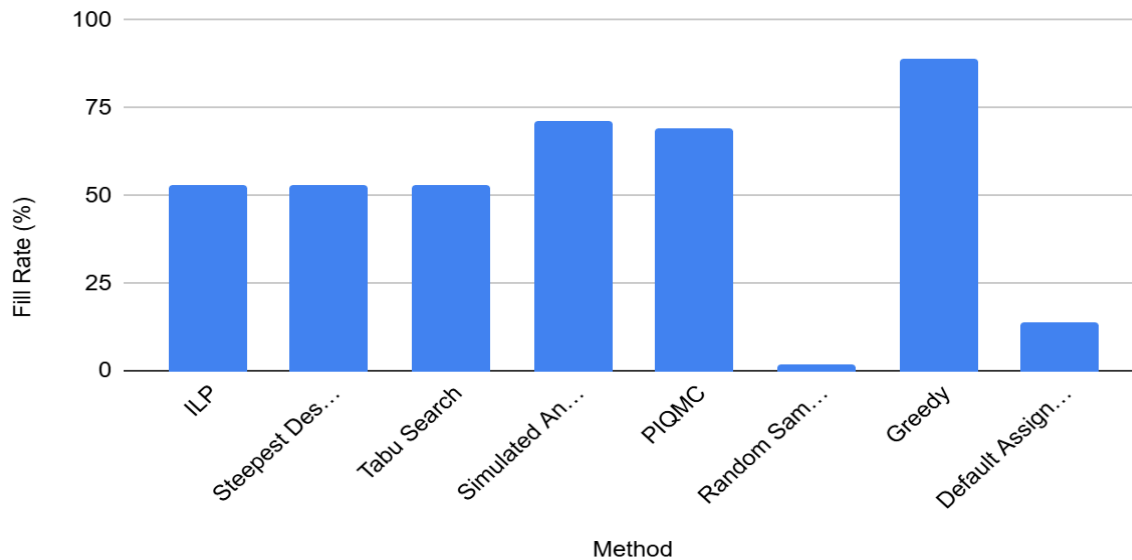


Figure 10. Fill rate comparison across optimization approaches

The fill-rate results show that maximizing the overall objective value does not necessarily maximize the number of fulfilled orders. For example, the Greedy approach achieved a higher fill rate than ILP, while its objective value was lower. This occurs because the objective function considers order value, shipping cost, and unfulfilled-order penalties rather than fill rate alone.

Runtime Comparison

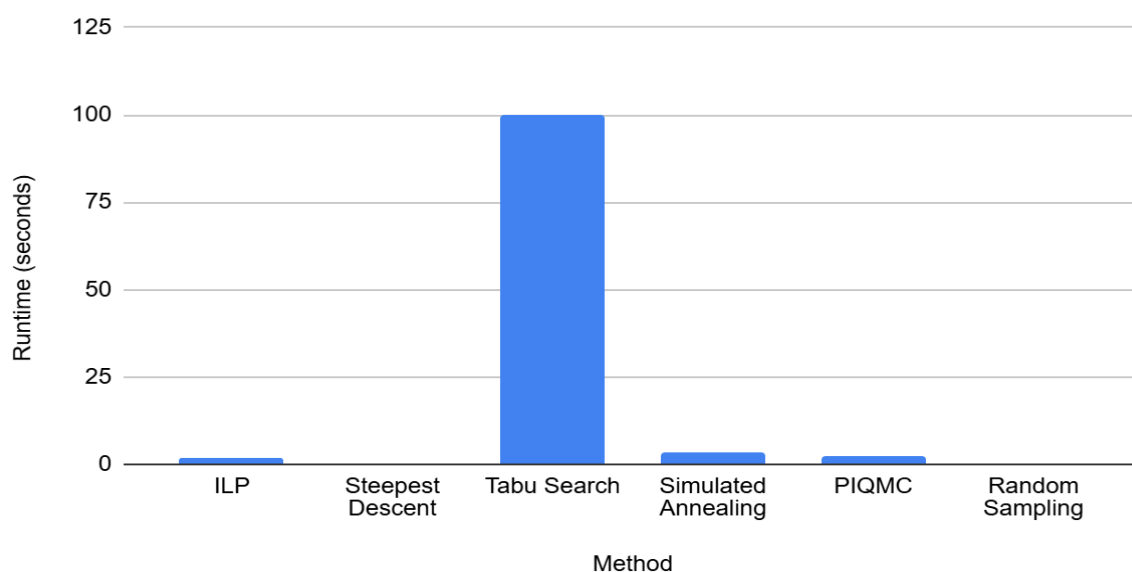


Figure 11. Runtime comparison across optimization approaches

The runtime comparison shows substantial differences in computational cost across the methods. Steepest Descent required approximately 0.08 seconds, while Tabu Search required approximately 100.2 seconds for the evaluated instance. Random Sampling had the lowest recorded runtime at approximately 0.003 seconds. These results illustrate the trade-off between search complexity and solution quality.

6.2 QAOA Results

The QAOA experiment was evaluated on the small six-variable instance containing 3 orders and 2 distribution centers. The experiment used 6 qubits and a QAOA depth of $p=2$. Since the instance was small enough to enumerate exhaustively, a brute-force search was used to determine the exact optimum for comparison.

Table 6. QAOA results compared with the exact brute-force solution

Metric	Objective Value	Fill Rate
Brute-force optimum	256,412	67%
QAOA	191,974	33%

The brute-force search obtained an objective value of approximately **256,412**, while QAOA obtained approximately **191,974**. The corresponding fill rates were **67%** for the brute-force solution and **33%** for the QAOA solution. The QAOA result had an optimality gap of approximately **25.1%** relative to the brute-force optimum.

These results indicate that the QAOA implementation was able to generate a candidate solution for the small DOM instance, but the experiment did not reach the exact optimum. Therefore, the current experiment demonstrates the application of QAOA to the formulated optimization problem rather than demonstrating quantum advantage.

The comparison between the exact brute-force solution and the QAOA result is shown in Figure 8.

6.3 Objective Value and Fill Rate

Objective value and fill rate measure different aspects of the fulfillment solution. The fill rate represents the percentage of orders that are successfully fulfilled, while the objective value represents the overall business value of the solution after considering order value, shipping cost, and penalties for unfulfilled orders. Therefore, a higher fill rate does not necessarily result in a higher objective value.

For example, the Greedy approach achieved a fill rate of **89%**, which was higher than the **53%** achieved by ILP. However, the objective value of Greedy was **412,060**, compared with **447,822** for ILP. Similarly, Simulated Annealing achieved a fill rate of **71%** but obtained an objective value of **369,752**. These results demonstrate that the optimization objective is not simply to maximize the number of fulfilled orders.

Objective Value vs Fill Rate (%)

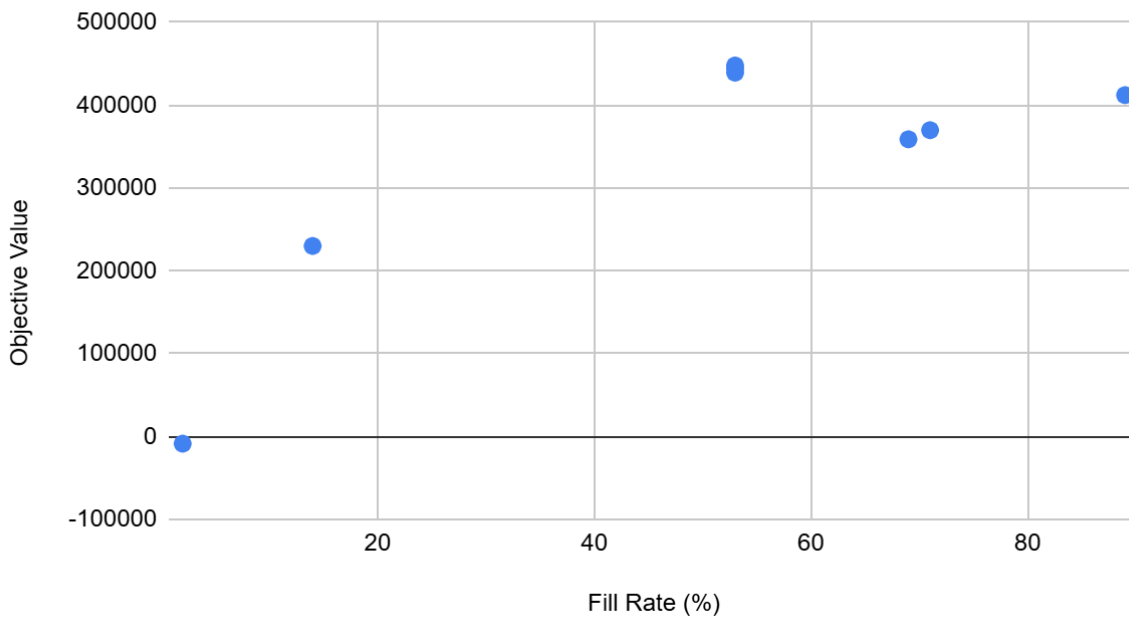


Figure 12. Relationship between objective value and fulfillment rate

The comparison shows that fulfillment rate should not be interpreted independently of the optimization objective. A solution may fulfill more orders while generating lower overall business value because additional assignments can introduce higher transportation costs or lower-value fulfillment decisions. The optimization model therefore evaluates the combined economic impact of fulfillment decisions.

6.4 Runtime Comparison

Computational runtime provides an indication of the processing cost associated with each optimization approach. The recorded runtimes varied considerably across the evaluated methods. Steepest Descent required approximately **0.08 seconds**, while Tabu Search required approximately **100.2 seconds** for the same optimization instance.

Random Sampling had the lowest recorded runtime at approximately **0.003 seconds**, but its objective value was substantially lower than the ILP reference. Simulated Annealing and PIQMC required approximately **3.2 seconds** and **2.2 seconds**, respectively. The ILP solution was obtained in less than **2 seconds**.

These results illustrate the trade-off between computational cost and solution quality. A faster method does not necessarily produce a better optimization result, while more extensive search strategies may require additional computation. The runtime comparison for the evaluated approaches is shown in **Figure 11**.

6.5 Comparison with the Provided DOM Reference Output

As an additional validation step, the provided order-level DOM output was examined separately from the reduced optimization experiment. The reference output contains 1,109 order-level records. Among these records, 1,106 were marked as Default and 3 were marked as Non-Default.

The three Non-Default recommendations correspond to orders 5484504521, 5484650344, and 5484739038. Their reported fill-rate improvements at the recommended distribution centers were approximately 25.0%, 14.6%, and 7.5%, respectively.

All three recommendations therefore satisfy the documented 5% fill-rate improvement criterion from the provided DOM formulation.

This reference-output analysis is presented separately from the 93-order optimization experiment. The reference output represents the provided business-level DOM result, whereas the optimization experiment uses a reduced mathematical formulation to compare different optimization approaches.

7. SCALABILITY ANALYSIS

The size of a distributed order management optimization problem increases rapidly as the number of orders and distribution centers grows. In the binary formulation, each possible order-to-distribution-center assignment introduces a decision variable. Therefore, increasing the number of orders or distribution centers directly increases the size of the optimization problem.

7.1 Problem Size Growth

In the main reduced formulation, 93 orders and 8 distribution centers resulted in **744 binary variables**. However, the complete focus-order set contains 359 orders, which would result in **2,872 variables** when considered across the same 8 distribution centers. Considering all 614 orders would result in **4,912 variables**. A full multi-SKU and multi-date formulation would increase the problem size substantially further.

Table 7. Growth of the optimization problem size

Problem Scope	Orders	Distribution Centers	Binary Variables
QAOA experiment	3	2	6
Reduced optimization	93	8	744
All focus orders	359	8	2,872
All open orders	614	8	4,912
Full multi-SKU/date formulation	—	—	~2,000,000+

Optimization Problem Size Growth

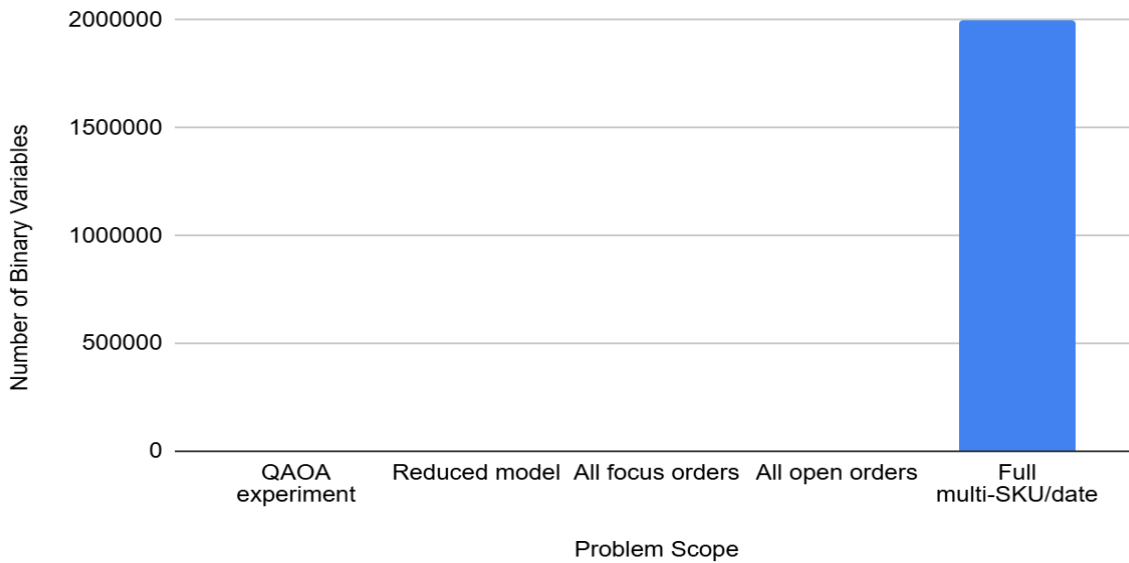


Figure 13. Growth in the number of binary decision variables with problem size

7.2 Implications for Quantum Optimization

The scalability analysis highlights the main challenge in applying quantum optimization to the distributed order management problem. Although the reduced model contains 744 binary variables, the QAOA experiment in this project was limited to a six-variable instance because quantum circuit simulation becomes increasingly expensive as the number of qubits grows.

The increase from 6 variables in the QAOA experiment to 744 variables in the reduced optimization model demonstrates the gap between a small quantum experiment and a larger real-world optimization problem. Extending QAOA to larger instances would require more qubits, improved circuit construction, efficient parameter optimization, and suitable quantum hardware or hybrid decomposition techniques.

Therefore, the current quantum experiment should be viewed as a proof-of-concept for applying QAOA to the DOM formulation rather than as a solution to the full-scale problem.

8. ASSUMPTIONS AND LIMITATIONS

The optimization study uses several assumptions and simplifications to make the distributed order management problem computationally tractable and suitable for experimentation. These assumptions should be considered when interpreting the results.

8.1 Assumptions

The main assumptions used in the project are:

- The optimization model focuses on **single-SKU orders** for the main 93-order experiment.
- The considered distribution centers are treated as available fulfillment locations according to the prepared input data.
- Shipping costs are represented using the available order-to-distribution-center cost information.
- Operational capacity is represented using simplified capacity constraints.
- Each order can be assigned to at most one distribution center.
- The objective function combines order value, transportation cost, and the impact of unfulfilled orders.
- The QAOA experiment uses a small subset of the complete optimization problem.

The detailed operational constraints in the provided DOM formulation, including case-pick, pallet-pick, dock availability, throughput, lead-time, and detailed penalty activation rules, were simplified or approximated in the optimization model.

8.2 Limitations

The project also has several limitations. The main optimization experiment uses 93 single-SKU focus orders rather than the complete multi-SKU order dataset. Some operational constraints, such as detailed throughput, dock, case-pick, and pallet-pick requirements, were simplified into capacity representations.

The penalty and diversion logic used in the optimization model is an approximation of the business requirements rather than an exact reproduction of a production system. In addition, the QUBO formulation uses penalty terms to represent constraints, and the selected penalty formulation may not capture every operational condition exactly.

The QAOA experiment was limited to **6 qubits** and was executed using the **Qiskit Aer simulator**. It was therefore not performed on physical quantum hardware and does not demonstrate quantum advantage. The quantum-inspired and stochastic optimization methods may also produce different results depending on their initialization and random seed.

The project therefore does not represent a complete production-scale reproduction of the detailed DOM business logic. In particular, the simplified model does not capture every operational rule simultaneously.

Table 8. Summary of project assumptions and limitations

Area	Project Limitation
Dataset	93 single-SKU orders used for main model
Operational constraints	Simplified capacity representation
Business rules	Penalty/diversion logic approximated
QUBO	Constraints represented through penalty terms
QAOA	Limited to 6 qubits
Hardware	Simulator; no physical quantum hardware
Stochastic methods	Results can vary with initialization/seed

9. KEY FINDINGS AND DISCUSSION

The experimental results provide a comparison of classical, heuristic, quantum-inspired, and quantum optimization approaches for the distributed order management problem. The results also highlight the difference between obtaining a high objective value and achieving a high fulfillment rate.

9.1 Classical Optimization Results

The ILP approach produced an objective value of approximately **447,822** and was used as the reference solution for the reduced 93-order problem. Steepest Descent produced an objective value of approximately **444,071**, which was close to the ILP result while requiring approximately **0.08 seconds**. Tabu Search also produced a relatively close objective value of approximately **439,188**, but required approximately **100.2 seconds**.

These results show that different search strategies can produce solutions with different combinations of solution quality and computational cost. The ILP model provides the reference solution, while heuristic methods provide alternative approaches with different runtime characteristics.

9.2 Objective Value and Fulfillment Trade-off

The experiments demonstrate that objective value and fill rate represent different performance measures. The Greedy approach achieved the highest recorded fill rate of **89%**, but its objective value of **412,060** was lower than the ILP objective value of **447,822**. This difference occurs because the objective function considers order value, shipping cost, and unfulfilled-order penalties rather than the number of fulfilled orders alone.

Therefore, evaluating an optimization method using only fill rate would not fully represent the business objective of the DOM problem.

9.3 Quantum Optimization Findings

The QAOA experiment demonstrated the application of a quantum optimization workflow to a small DOM instance. The experiment used **3 orders, 2 distribution centers, 6 binary variables, and 6 qubits** with a QAOA depth of $p = 2$.

The brute-force optimum for this small instance was approximately **256,412**, while QAOA obtained approximately **191,974**, resulting in an optimality gap of approximately **25.1%**. Thus, QAOA generated a candidate solution but did not reach the exact optimum in this experiment.

The experiment therefore demonstrates the feasibility of formulating the DOM problem for QAOA, but it does **not establish quantum advantage**. Larger experiments and execution on suitable quantum hardware would be required to investigate the potential benefits of quantum optimization at scale.

9.4 Overall Discussion

Overall, the project shows that distributed order management can be formulated as a binary optimization problem and transformed into QUBO and Ising representations for quantum optimization. Classical and quantum-inspired methods provide useful benchmarks for evaluating the optimization problem, while the QAOA experiment demonstrates how the same formulation can be explored using a quantum algorithm.

The main challenge is scalability. The reduced model contains **744 binary variables**, whereas the QAOA experiment was limited to **6 variables**. This difference highlights the need for improved quantum algorithms, larger quantum systems, and hybrid optimization strategies before applying QAOA directly to larger real-world DOM instances.

10. FUTURE SCOPE

10.1 Scaling the Quantum Approach

A major direction for future work is to extend the QAOA experiment from the current six-variable instance to larger optimization problems. This would involve studying methods for reducing the number of required qubits, decomposing large problems into smaller subproblems, and efficiently mapping larger QUBO formulations to quantum circuits.

10.2 Improved QAOA Performance

Future experiments can investigate improved QAOA performance through better parameter initialization, alternative classical optimizers, increased circuit depth, and different mixer designs. These improvements can be evaluated by comparing the resulting solutions with exact classical solutions on instances where exact evaluation is still computationally feasible.

10.3 Real Quantum Hardware

The current QAOA experiment was performed using the Qiskit Aer simulator. Future work can evaluate the formulation on available quantum hardware to study the effects of hardware noise, limited connectivity, gate errors, and circuit depth. Error-mitigation techniques could also be investigated to improve the quality of measured solutions.

10.4 Larger and More Realistic DOM Models

The current model focuses on a reduced set of single-SKU orders. Future work can incorporate multi-SKU orders, multiple planning dates, detailed inventory availability, transportation constraints, warehouse operating hours, and more detailed throughput requirements. This would provide a closer representation of real-world distributed order management.

10.5 Hybrid Optimization Strategies

Hybrid approaches combining classical optimization, quantum-inspired algorithms, and QAOA can also be explored. For example, classical methods could be used to reduce or decompose a large problem before applying QAOA to selected subproblems. Solutions generated by quantum optimization could then be refined using classical local-search methods.

Overall, future development should focus on increasing problem size, improving QAOA solution quality, incorporating realistic operational constraints, and evaluating hybrid quantum-classical approaches. These extensions would help determine the practical applicability of quantum optimization to larger distributed order management problems.

11. CONCLUSION

This project formulated Distributed Order Management as a binary optimization problem in which customer orders are assigned to suitable distribution centers while considering order value, shipping cost, inventory availability, and operational capacity. The reduced optimization model considered **93 single-SKU focus orders across 8 distribution centers**, resulting in **744 binary decision variables**.

The formulated problem was evaluated using several classical and quantum-inspired optimization approaches. **ILP** provided the reference solution with an objective value of approximately **447,822**. Heuristic methods such as Steepest Descent and Tabu Search produced solutions close to the ILP reference, while other approaches demonstrated different trade-offs between objective value, fulfillment rate, and runtime.

The optimization formulation was subsequently transformed into **QUBO and Ising representations** to investigate quantum optimization using **QAOA**. Due to the computational requirements of quantum simulation, QAOA was evaluated on a small instance containing **3 orders, 2 distribution centers, and 6 qubits**. The QAOA solution achieved an objective value of approximately **191,974**, compared with the brute-force optimum of approximately **256,412**, resulting in an optimality gap of **25.1%**.

The results demonstrate that the DOM problem can be represented in a form suitable for quantum optimization and that QAOA can be experimentally evaluated on a small instance. However, the current experiment does not demonstrate quantum advantage, and the gap between the six-qubit experiment and the **744-variable** reduced model highlights the scalability challenge.

Overall, the project establishes a complete workflow from business problem formulation to classical optimization, QUBO and Ising transformation, and small-scale quantum optimization. Future work can focus on larger quantum experiments, improved QAOA strategies, realistic operational constraints, quantum hardware, and hybrid quantum-classical optimization methods.

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